

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026

Homework #9: due **Wednesday**, April 29th, at **5PM**.

Show all work for full credit!

Submit your work as a single PDF on Gradescope, labeling each problem number with a page.

This final homework is a little different; you need to use some numerical computation on actual data (using Python, Matlab, Excel, Google Sheets, etc.) and your knowledge of probability to answer the questions. The data you need is in the file <https://sites.ecse.rpi.edu/~rjradke/hw9.zip>.

1. (35 points.) First we're going to revisit the Hades/Silksong problem from HW7, except you just have the game-playing records given in the file `hw9p1.txt`, encoded as follows:

- Column 1: whether the player was playing Hades 2 (coded as a 1) or Silksong (coded as a 2)
- Column 2: the number of player deaths per hour, which we hypothesize is drawn from a Gaussian distribution with unknown parameters (μ_i, σ_i)
- Column 3: the total hours played so far, which we hypothesize is drawn from an exponential distribution with unknown parameter λ_i
- Column 4: whether the player has broken their controller in frustration, which we hypothesize is drawn from a Bernoulli distribution with unknown parameter p_i

- (a) (5 points.) Estimate the prior probabilities $P(H)$ and $P(S)$.
- (b) (5 points.) Estimate the class-conditional parameters of the Gaussian distributions (μ_H, σ_H) and (μ_S, σ_S) . (Use the maximum likelihood estimate for μ , and the version of the σ estimate where we divide by n , not $n - 1$.)
- (c) (5 points.) Estimate the class-conditional parameters of the exponential distributions λ_H and λ_S .
- (d) (5 points.) Estimate the class-conditional parameters of the Bernoulli distributions p_H and p_S .
- (e) (5 points.) We now observe that a particular player has died an average of 2.03 times per hour, played 1 hour so far, and has **not** broken their controller in frustration. Based on all of your parameter estimates, what game do we decide the gamer is playing if we apply the Maximum Likelihood Decision Rule?
- (f) (5 points.) What game do we decide the gamer is playing if we apply the Maximum a Posteriori (MAP) Decision Rule?
- (g) (5 points.) Finally, suppose that we are extremely eager to find Hades players and are worried about missing any, so we set the costs $C_{SS} = C_{HH} = 0$, $C_{HS} = 2$, $C_{SH} = 1$ and apply the Bayesian Decision Rule that includes costs. What game would we decide now?

More fun on the next page →

2. (30 points) The data in the file `hw9p2.txt` represents 8000 observations of a random variable that can take on the values 0 to 7.

- (a) (5 points) Plot the histogram of the data and determine its sample mean.
- (b) (5 points) Compute the test statistic D^2 for the hypothesis that the observations come from a uniform distribution over the integers $\{0, 1, \dots, 7\}$.
- (c) (5 points) Can we reject the hypothesis at the 5% significance level? How about at the 1% significance level? Explain your answers with reference to Table 8.3 on p. 438 (or some other chi-squared table). Clearly justify your answer.
- (d) (10 points) Now compute the test statistic D^2 for the hypothesis that the observations come from a Poisson distribution with mean 2. Since the Poisson distribution has infinitely many outcomes, you should take the probability for outcomes above $n = 7$ and add that to the bin $n = 7$ so that you have a PMF that sums to 1.
- (e) (5 points) Can we reject the hypothesis at the 5% significance level? How about at the 1% significance level? Again, justify your answers using a chi-squared table.

3. (10 points.) The data in the file `hw9p3.txt` represents 10000 observations of X , a continuous uniform random variable on $[-2, 2]$.

- (a) (7 points.) Determine the transformation $Y = f(X)$ that results in observations of a random variable with PDF

$$f_Y(y) = \begin{cases} \frac{3}{8}y^2 & y \in [0, 2] \\ 0 & \text{otherwise} \end{cases}$$

- (b) (3 points.) Apply your transformation to the data in the file and plot a histogram of the result to verify it indeed has the correct PDF.

4. (25 points.) The data in the file `hw9p4.txt` encodes many observations of the joint random variable (X, Y) . In this problem, we will compute estimators for X given Y . That is, we will determine functions of the form $\hat{X} = g(Y)$ so that given a value of Y , we produce an estimate $\hat{X}$ of X . Since Y is discrete, your estimators should be in the form

$$\hat{X} = \begin{cases} \dots & Y = 0 \\ \dots & Y = 1 \\ \dots & Y = 2 \end{cases}$$

- (a) (5 points.) Based on the data in the file, construct an estimate of the joint PMF of X and Y (include this 3x3 table in your homework writeup).
- (b) (5 points.) Compute the ML estimator of X given Y .
- (c) (5 points.) Compute the MAP estimator of X given Y .
- (d) (5 points.) Compute the probability of error for these two estimators.
- (e) (5 points.) Compute the MMSE estimator of X given Y . (Unlike in the previous parts, the answer will not have integer values for $\hat{X}$.)